



United International University

Department of Computer Science and Engineering

CSE 1115: Object Oriented Programming

Midterm Exam Trimester: Spring 2025 Time: 1 hour 30 minutes Marks: 30

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions.

1. a) Consider the following codes and write the output

[5 Marks] [CO1]

MindGame.java

```
class MindGame {
    int a = 0;
    int b = 2;
    static int c = 0 ;
    static int d = 5;

    MindGame()
    {
        a--;
        b += 5;
        c++;
        d--;
    }

    static {
        c--;
        d += c;
    }

    {
        a = c++;
        b = b+d;
    }

    public void displayAB()
    {
        System.out.println("a = "+a);
        System.out.println("b = "+b);
    }

    public static void displayCD()
    {
        System.out.println("c = "+c);
        System.out.println("d = "+d);
    }
}
```

Handwritten notes for MindGame.java:

- ③ a = -2, b = 11, c = 1, d = 3
- ① c = -1, d = 4
- ② a = -1, b = 6, c = 0

Main.java

```
public class Main {
    public static void main(String[]
        args)

    {
        MindGame.displayCD();
        MindGame m1 = new MindGame();
        m1.displayAB();
        m1.displayCD();
        MindGame.displayCD();
    }
}
```

(b) Main.java program provided below which contains a main method that creates multiple instances of the Shape class using different constructors. Your task is to complete Shape.java by adding necessary constructors so that Main.java runs correctly.

[5 Marks] [CO1]

Main.java

```
public class Main {
    public static void main(String[] args) {
        // set default value 10 to width, height and length
        Shape shape1 = new Shape();

        //set width = 5, height = 10, length = 0
        Shape shape2 = new Shape(5, 10);
        //set width = 3, height = 6, length = 9
        Shape shape3 = new Shape(3, 6, 9);

        // set width, height and length to shape3's width, height and length respectively.
        Shape shape4 = new Shape(shape3);
    }
}
```

Shape.java

```
class Shape {
    int length, width, height;
    // add necessary constructors
}
```

2. Consider the following class named **Burger** representing a generic burger. Each burger should have a name and a size as attributes. The **Burger** class has a prepare() method that always prints a base message with the burger's name and size.

Now, you have two subclasses of burgers: **BeefBurger** and **VeggieBurger**. Each type has two additional attributes (field) named flavor (data type: **String**) and price (data type: **int**). Both burgers have a prepare() method which prints the base message at first and then prints the additional details (flavor and price).

Based on the given code and expected output, write the necessary code so that the expected output is produced. Note that you **cannot change** any given code. If you **complete the code and run**, then the following output should be found:

[3+3=6 Marks] [CO1]

Output:

Preparing Double Decker Large burger

Flavor: Spicy

Price: 400

Preparing Mushroom Delight Small burger

Flavor: Sweet

Price: 300


```

class Burger {
    private String name;
    private String size;

    public Burger(String name, String size) {
        this.name = name;
        this.size = size;
    }
    // always print the base message
    public void prepare() {
        System.out.println("Preparing " + name + " " + size + " burger");
    }
}

class MainBurger {
    public static void main(String[] args) {

        Burger beefBurger = new BeefBurger ("Double Decker", "Large", "Spicy", 400);
        beefBurger.prepare();

        Burger veggieBurger = new VeggieBurger ("Mushroom Delight", "Small", "Sweet",
            300);
        veggieBurger.prepare();
    }
}

```

3. The given Java code defines a **Vehicle** class with two methods: *start()* and *move()*. The **Bus** and **Cycle** classes inherit from **Vehicle**. The **VehicleTest** class implements the *main()* method. Follow the instructions. [2+2+2=6 Marks] [CO1]

- Identify and correct errors in the given Java code.
- You cannot modify the **Vehicle** class.
- You cannot add new methods to any class.
- Add necessary correction to the *main()* method and the sub classes (Bus and Cycle).
- Provide the output of the corrected code.

```

public class Vehicle {

    void start(){
        System.out.println("Vehicle is
            starting");
    }

    final void move(){
        System.out.println("Vehicle is
            moving");
    }
}

```

```

class Bus extends Vehicle{
    void move(){
        System.out.println("Bus is moving");
    }

    void needFuel(){
        System.out.println("Bus needs fuel");
    }

    void start(){
        System.out.println("Bus is
            starting");
    }
}

```

```

class Cycle extends Vehicle {
    void pedal(){
        System.out.println("Pedal the
            cycle");
    }
}

```

```

class VehicleTest{

    public static void main(String[] args) {
        Vehicle bus = new Bus();
        bus.start();
        bus.needFuel();
        Vehicle cycle = new Cycle();
        cycle.move();
        cycle.pedal();
    }
}

```

4. (a) Consider the following code FlightBooking.java.

FlightBooking.java

```

public class FlightBooking {
    private String passengerName;
    private int seatNumber;
    private boolean confirm = false;
    public FlightBooking(String passengerName, int seatNumber) {
        this.passengerName = passengerName;
        this.seatNumber = seatNumber;
    }
}

```

Based on the class, a main class FTest.java is written as follows:

FTest.java

```
class FTest{
    public static void main(String[] args) {
        FlightBooking f1 = new FlightBooking("John", 12);
        FlightBooking f2 = new FlightBooking("Maria", 3);
        if( f1.confirm == false ){
            f1.confirm = true;
            f1.seatNumber = 32;
        }
        System.out.println("seat number of Maria: " + f2.seatNumber );
        System.out.println("seat number of John: " + f1.seatNumber);
    }
}
```

Is there any error in *main()* method? If yes, modify the codes in "FlightBooking.java" and "FTest.java". Note that you **cannot** change access modifiers of any instance variables in FlightBooking.java.

[1+4 = 5 Marks] [CO1]

(b) Consider the following Box.java and Main.java. What is the output of the program? Justify your answer.

[3 Marks] [CO1]

```
public class Box {
    double width; = 1
    double height; = 2
    double depth; = 3

    public void print(){
        System.out.println(width);
        System.out.println(height);
        System.out.println(depth);
    }
}
```

```
public class Main {
    public static void main(String[]
        args) {
        Box b1= new Box();
        b1.width=1;
        b1.height=2;
        b1.depth=3;
        Box b2 = b1;
        b1=null;
        b2.print();
    }
}
```